EX.NO: 03

DATE : 16/07/2026

STUDY OF DIFFERENT TYPES OF CABLES AND CRIMPING OF CABLE WITH RJ45 CONNECTOR

AIM :-

To study different types of network cables & understand the crimping process using an RJ45 Connector.

CONCEPT :-

Network cables are used to connect computers and other networking devices for data communication. The most commonly used cable in LAN is the unshielded Twisted Pair (UTP) cable. An RJ45 connector is attached to the ends of the cable of establish a network connection. The wires are arranged according to the T568A or T568B wiring standard before crimping, & the completed cable is tested to ensure proper connectivity.

TYPES OF NETWORK CABLES :-

CABLE TYPE	DESCRIPTION
1) UTP (unshielded Twisted pair)	commonly used for Ethernet LAN's and office networks.
2) STP (shielded Twisted pair)	Used in area with high electrical or electromagnetic interference.
3) Coaxial Cable	Used for cable television, broadband internet, & CCTV systems.
4) Optical Fiber Cable.	Used for high-speed, high bandwidth & long distance communication.

CRIMPING PROCEDURE :-

1) Cut the UTP cable to the required length.
2) Strip about 2-3 cm of the outer insulation.
3) Untwist and straighten the wires.
4) Arrange the wires according to the T568B standard.
5) Trim the wire ends evenly.
6) Insert the wires into the RJ45 connector.
7) Crimp the connector using a crimping tool.
8) Test the cable using a LAN cable tester.

OBSERVATION :-

(i) The wires were arranged according to the T568B Standard.

(ii) The RJ45 connector was crimped successfully.

(iii) The cable tester indicated proper continuity.

(iv) The cable was ready for network communication.

RESULT :-

16/7/25

The different types of cables were studied, & the UTP cable was successfully terminated using & RJ45 connector & tested for proper connectivity.